

Krzysztof Broniszewski

AI / Machine Learning / Computer Vision Engineer

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CV LinkedIn GitHub Portfolio

PROFESSIONAL PROFILE

I develop solutions in Machine Learning, Computer Vision and AI, primarily using Python, PyTorch, OpenCV and scikit-learn. I build end-to-end projects - from data preparation and analysis, through model training and evaluation, to applications that demonstrate their operation. I am particularly interested in Computer Vision, Generative AI and practical AI applications. I combine technical skills with many years of experience in graphics, digital publishing and production workflow optimization.

TECHNICAL SKILLS

Programming

Python • SQL

Machine Learning

scikit-learn • classification • regression • preprocessing • feature engineering • model evaluation

Deep Learning

PyTorch • TensorFlow/Keras • CNN • transfer learning

Computer Vision

OpenCV • YOLO • MediaPipe • image processing and analysis

Generative AI / NLP / LLM

ComfyUI · FLUX · Stable Diffusion / SDXL · ControlNet · LoRA · IP-Adapter · Inpainting · Upscaling · WAN Video
NLP: tokenization · lemmatization · sentiment analysis · embeddings · BERT / Transformers

Data & Visualization

Pandas • NumPy • Matplotlib • Plotly

Databases

PostgreSQL • BigQuery

Applications / Tools

Dash • React • Flask • FastAPI • Git • GitHub • Docker • Jupyter Notebook • Google Colab • Ollama • FAISS

PROJECTS

01 **RAV4 RAG Assistant - Generative AI / RAG / LLM**

End-to-end RAG assistant for technical questions based on Toyota RAV4 documentation. PDF processing, chunking, embeddings, semantic retrieval, reranking and local LLM generation.
Technologies: Python · RAG · E5 Embeddings · FAISS · BGE Reranker · Bielik 11B · Ollama · FastAPI GitHub

02 **Face Emotion Recognition - Deep Learning / Computer Vision**

Facial emotion recognition model using ResNet18 and transfer learning, including data pipeline, training, evaluation and image inference.
Technologies: Python · PyTorch · OpenCV · ResNet18 GitHub

03 **Drone Image Quality Assessment - Deep Learning**

Automatic quality assessment of drone imagery using deep learning, combining image analysis with practical Computer Vision for material selection.
Technologies: Deep Learning · Computer Vision · image quality analysis GitHub

04 **Road Accident Analysis in Poland 2004-2023**

Interactive dashboard based on Polish Police data, including data preparation, transformation and interactive time-based visualizations.
Technologies: Python · Pandas · Plotly · Dash GitHub

05 **California Housing - Machine Learning**

Regression project covering data preparation, feature analysis, model training, comparison and evaluation.
Technologies: Python · Pandas · scikit-learn GitHub

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PROFESSIONAL EXPERIENCE

Walstead Krakow Sp. z o.o. | Krakow

Production Costing and Preparation Specialist ISM | 02.2023-11.2026

- Prepared production cost estimates in Signa and analyzed costs, materials, supplier and subcontractor offers.
- Independently created and developed an Excel tool for monitoring orders, production stages, deadlines and missing data.
- Coordinated production processes, planned tasks and deadlines, and cooperated with clients, suppliers and internal teams.
- Performed technical verification and preparation of files and print impositions, ensuring compliance with specifications.
- Analyzed and improved workflows, solved technical problems and automated selected repetitive tasks.

Earlier Professional Experience | 2005-2023

DTP / Prepress / Digital Printing / Graphic Design

- Many years of experience in technical preparation and verification of graphic files, DTP, prepress, digital and offset production.
- Worked with advanced production preparation and imposition systems, Adobe software, CorelDRAW and file-control tools.
- Designed graphic materials and packaging and prepared them for different production technologies.
- Experience managing production, a small team, warehouse and procurement.
- Ran E-GRAPH sole proprietorship - digital production, graphic design, client service and business management.

Experience includes: Mellow • Belcaro/Pasaz • Brillo • Legra • Alnus • Drukarnia GS • Naswietlarnia B1+/Djaf

Self-employment: E-GRAPH

COURSES & CERTIFICATES

CodeBrainers - Data Science + AI Bootcamp

12.2024-06.2025 | 260 lesson hours

Python • SQL • Pandas • scikit-learn • XGBoost • Machine Learning • Deep Learning • CNN • NLP • Spark • Databricks • BigQuery

Practical Python | Python Programming Course

05.2022-07.2022 | 8 weeks | 8 projects

Python • Pandas • Jupyter Notebook • Flask • data analysis • Big Data • text and sentiment analysis • application development

EDUCATION

Krakow University of Technology

Faculty of Technical Physics and Computer Modelling

Unfinished studies | 2002-2004

High School in Muszyna

IT profile | 1996-2000

ACHIEVEMENTS

• Winner of the "We Can Do It! Starting Our Own Business" competition - prepared and defended a business plan and obtained full funding to launch E-GRAPH.

LANGUAGES

Polish - native

English - B1 (good command of technical documentation)

INTERESTS

Drones and aerial photography • Computer Vision • Generative AI • photography and post-production • video editing